Example 3.6

By proportionality law, check whether 3 : 2 and 30 : 20 are in proportion.

Solution

Here the extremes are 3 and 20 and the means are 2 and 30.

Product of extremes, ad = 3 × 20 = 60.

Product of means, bc = 2 × 30 = 60.

Thus by proportionality law, we find $ad = bc$ and hence 3 : 2 and 30 : 20 are in proportion.

Example 3.7

A picture is resized in a computer as shown below.

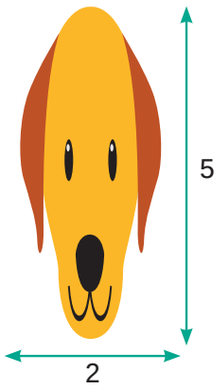


Fig. 3.6 (a)

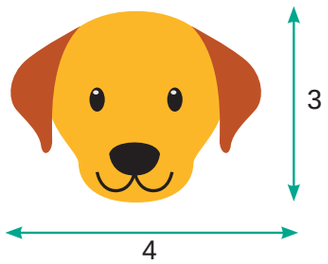


Fig. 3.6 (b)

TRY THIS

1. Fill the box by using cross product rule of two ratios $\frac{1}{8} = \frac{5}{\square}$
2. Use the digits 1 to 9 only once and write as many ratios that are in proportion as possible. (For example : $\frac{2}{4} = \frac{3}{6}$)

Do you observe any change in shape and size of the picture? Check whether the ratios formed by its length and breadth are in proportion by cross product method.

Solution

The given pictures are in the ratio 2 : 5 and 4 : 3 respectively.

Here the extremes are 2 and 3 and the means are 5 and 4.

Product of extremes, ad = 2 × 3 = 6.

Product of means, bc = 5 × 4 = 20.

Thus, we find $ad \neq bc$ and hence 2 : 5 and 4 : 3 are not in proportion.

3.4 Unitary Method

Finding the value of one unit and then using it to find the value of the required number of units is known as unitary method.

Steps involved in Unitary Method

- Express the given problem in Mathematical statement.
- Find the value of one unit of the given item using division.
- Find the value of the required number of the same items using multiplication.

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